

Question:
Four days after undergoing an elective total hip replacement, a 65-year-old woman develops a DVT that embolizes to the lung. Along with tachypnea, tachycardia, and cough, the patient would most likely present with a PaO2 of what?Answer Choices:A. 120 mmHgB. 110 mmHgC. 85 mmHgD. 60 mmHg

Reasoning + Answer:

<think>###FindingReasoningPaths : 1 . ** Surgicalevent (totalhipreplacement) â Postoperativeperiodâ Prolongedimmobilityâ Venousstasisâ Deepveinthrombosis (DVT)â Pulmonaryembolism (PE). ** 2 . ** Postoperativeperiodâ Orthopedicprocedureâ Compressionofveinsâ Venousstasisâ Deepveinthrombosis (DVT)â Pulmonaryembolism (PE). ** 3 . ** Electivetotalhipreplacementâ Surgicalriskfactorsâ Prolongedimmobilizationâ Venousstasisâ Deepveinthrombosis (DVT)â Pulmonaryembolism (PE). ** 4 . ** Venousstasisâ Deepveinthrombosis (DVT)â Pulmonaryembolism (PE). ** ---###ReasoningProcess : 1 . ** Understandingtheclinicalscenario : ** - Thepatientis a 65-year-oldfemalewhounderwentanelectivetotalhipreplacementfourdaysago . Postoperatively , shedevelopeda DVT (deepveinthrombosis) thatembolizedtothelung , resultinginapulmonaryembolism (PE). Thisisaknowncomplicationoforthopedicsurgeries , particularlywhenprolongedmmobilizationoccurs . 2 . ** KeyfactorscontributingtoDVTandPE : ** - ** Surgicalevent: ** Thetotalhipreplacementis a significantorthopedicprocedure , whichinherentlyincreasestheriskofDVT . Theimmobilityandpositioningofthepatientduringsurgery , particularlyinalateralposition , cancompresstheveins , leadingtovenousstasisandthrombusformation . - ** Postoperativeperiod: ** Thepatientislikelyinabedrestorlimitedmobilityforseveraldays , furthercontributingtovenousstasisduetoreducedvenousflow . - ** Orthopedicprocedure: ** Theimmobilityisawell-documentedriskfactorforDVT , asitdeprivesveinsofbloodflowandpromotesclotformation . 3 . ** PathophysiologyofDVTandPE : ** - Venousstasisleadsthrombusformationinthedeepveinsofthelegs (e.g . , femoral , popliteal , anddeepveinsofthecalf) . Iftheclotbreaksoff (embolizes) , itcantraveltothelungsviathebloodstream , causingapulmonaryembolism . - Pulmonaryembolismtypicallypresentswith**symptomsofhypoxia** (e.g . , shortnessofbreath , tachypnea , hypoxemia) becausetheembolusobstructspulmonarybloodflow , leadingtoimpairedoxygenationoftheblood . 4 . ** PaO2inthecontextofPE : ** - Pulmonaryembolismleadstodecreasedoxygensaturationduetoimpairedgasexchangeinthelungs . ThePaO2 (arterialpartialpressureofoxygen) valueduringapulmonaryembolismreflectsthereducedoxygen-carryingcapacityofthebloodduetotheobstruction . - APaO2typicallyfallsbelownormal (90mmHg) inthepresenceofsignificanthemodynamicorpulmonaryperfusionimpairmentcausedbyaPE . ---###Conclusion: ThePaO2inthiscasewouldmostlikelybe** 60mmHg ** , indicatingsignificantthemodynamidmpairmentandhypoxemiaduetothepulmonaryembolism . </think><answer> 60mmHg </answer>

